

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study ¹
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1 Objective

- To implement a Decision Tree classifier for breast cancer diagnosis.
- To extend the Decision Tree into a Random Forest ensemble model.
- To study the impact of hyperparameters on overfitting and generalization.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare the single-tree model and the ensemble-tree model on held-out test data.

2 Problem Statement

Given 30 numerical features computed from digitized images of fine needle aspirates (FNA) of breast masses, the task is to build a binary classifier that predicts whether a tumor is **Malignant (M)** or **Benign (B)**. A Decision Tree classifier and a Random Forest ensemble classifier are trained, tuned via 5-fold cross-validation on an 80% training split, and compared on a held-out 20% test split using Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC Curve and ROC-AUC.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
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Dataset Source	UCI Machine Learning Repository (donated by Dr. W.H. Wolberg, W.N. Street, O.L. Mangasarian, University of Wisconsin)
Number of Samples	569
Number of Features	30 real-valued numerical features (10 base cell-nucleus measurements $\times$ mean, standard error, and worst), after excluding the non-predictive id column
Number of Classes	2 (Benign = 0, Malignant = 1)
Class Balance	357 Benign (62.7%), 212 Malignant (37.3%)
Missing Values	0 (none)
Duplicate Rows	0 (none)
Train-Test Split	80% train (455 samples) / 20% test (114 samples), stratified, random_state=42

4 Exploratory Data Analysis

Class Distribution

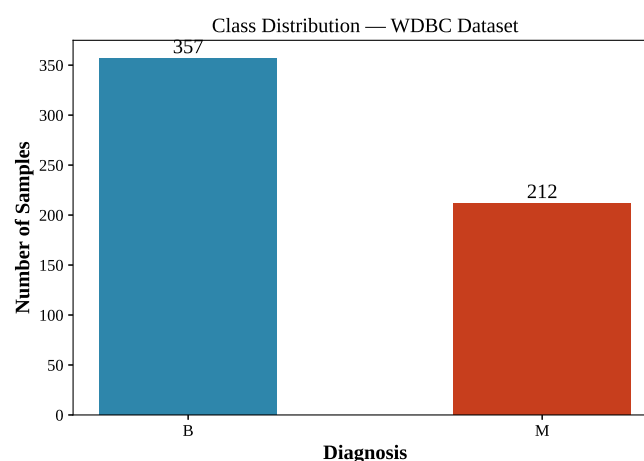


Figure 1: Class distribution of the WDBC dataset: 357 Benign vs 212 Malignant samples.

Inference: The dataset shows a moderate class imbalance, with benign cases outnumbering malignant cases by a ratio of roughly 1.68:1. This imbalance is not severe enough to require resampling techniques, but it does justify reporting F1-score and ROC-AUC alongside accuracy, since accuracy alone can be misleading on imbalanced binary classification tasks. Stratified splitting and stratified 5-fold cross-validation were therefore used throughout the experiment to preserve this 62.7:37.3 ratio in every train/validation partition.

Correlation Matrix

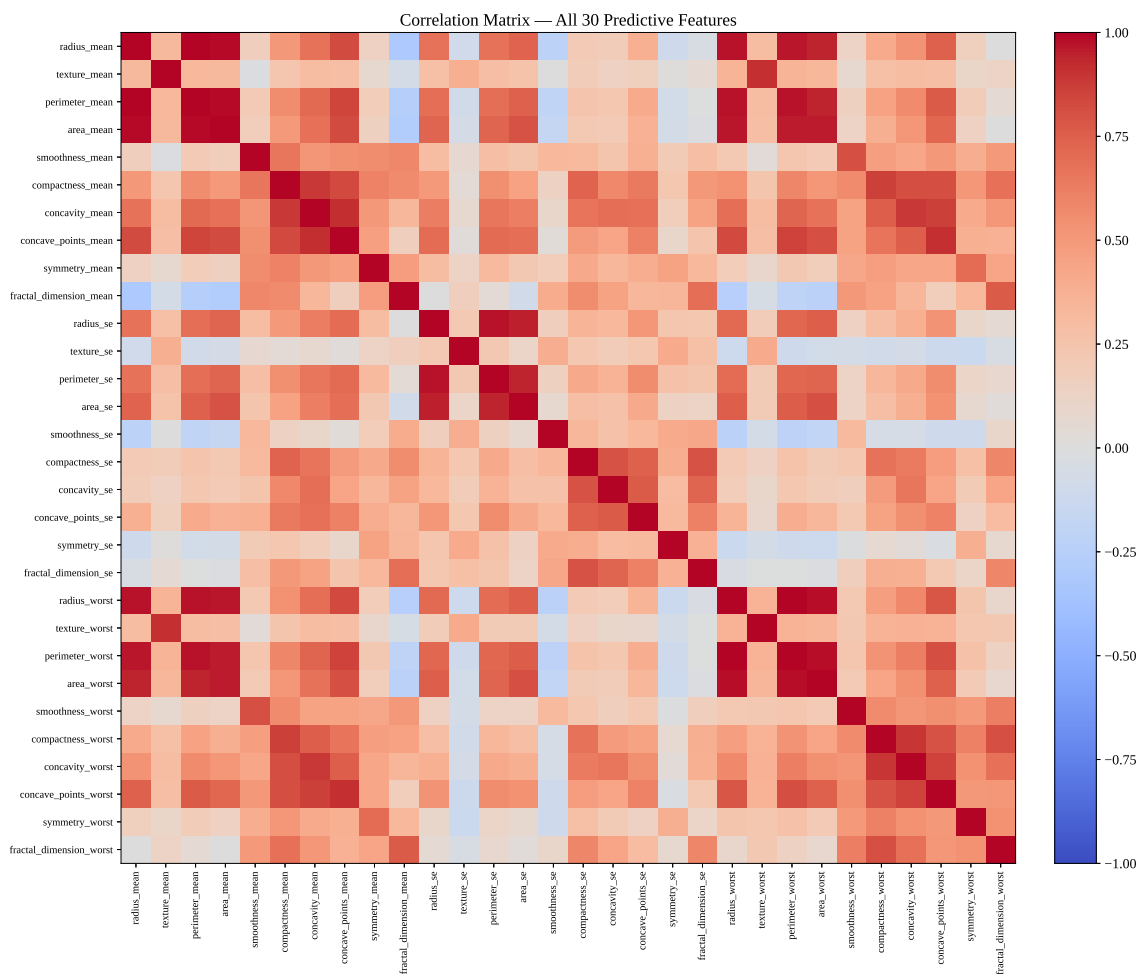


Figure 2: Correlation matrix of all 30 predictive features in the WDBC dataset.

Inference: Strong positive correlations (> 0.9) are visible among the size-related features, e.g. radius, perimeter, and area (for the mean, se, and worst statistics groups), which is expected since these are geometrically related measurements of the same cell nuclei. The concave_points, concavity,

and compactness features also correlate strongly with one another. In contrast, `fractal_dimension_mean` shows weak or mildly negative correlation with the size-based features. This multicollinearity is a key reason tree based models (which split on one feature at a time and are insensitive to feature scaling/redundancy) are well suited to this dataset.

Consolidated 12-Subplot EDA Summary

Exploratory Data Analysis — Wisconsin Diagnostic Breast Cancer Dataset

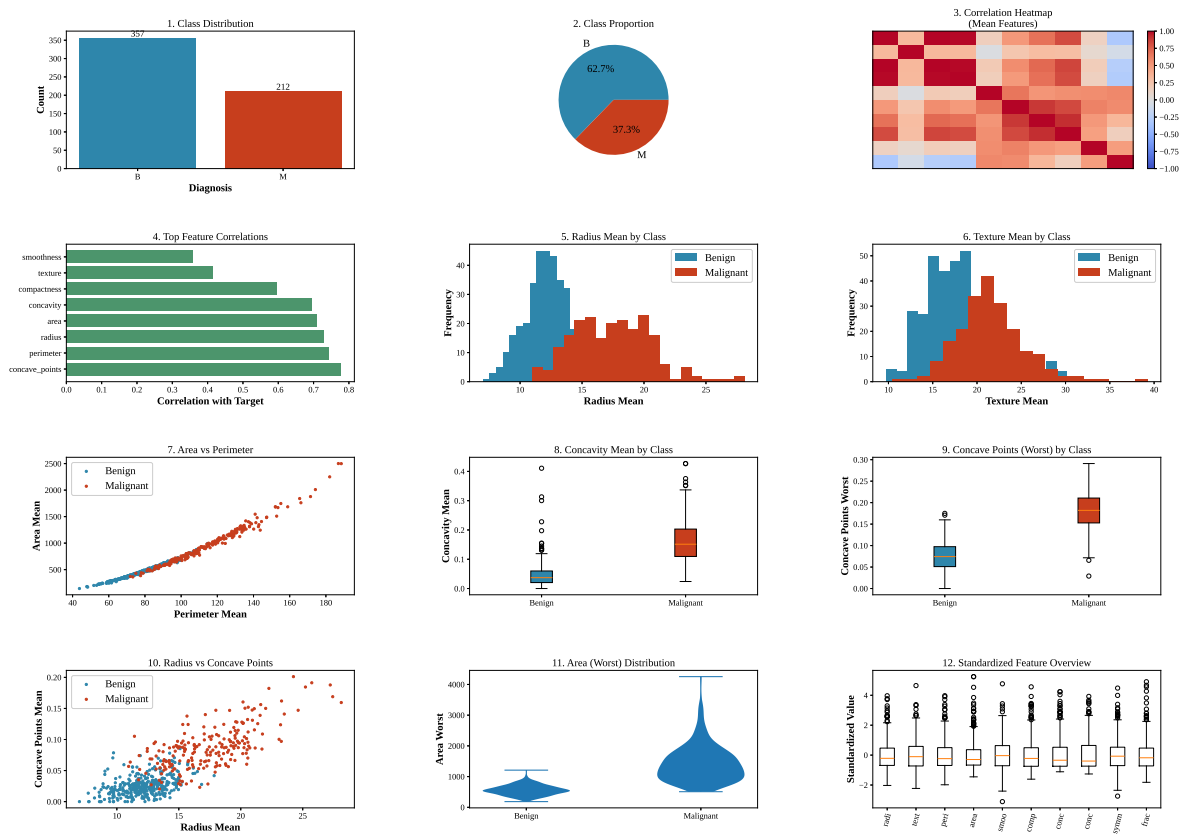


Figure 3: Consolidated 12-panel exploratory data analysis of the WDBC dataset, generated by the reusable `generate_eda_summary()` function.

Inference: The consolidated view confirms the class imbalance (panels 1-2), the strong correlation structure among size-based features (panel 3), and that `concave_points`, `concavity`, `area` and `perimeter` are the mean features most correlated with malignancy (panel 4). The distribution panels (5, 6, 8, 9, 11) show that malignant tumors tend to have systematically higher radius, texture, concavity, and worst-area values with wider spread than benign tumors, and the scatter panels (7, 10) show near-linear geometric relationships (area vs. perimeter) as well as reasonably good visual class

separability along radius and concave-points axes. Panel 12 shows that most standardized mean features contain a number of high-value outliers consistent with a subset of aggressively malignant samples.

5 Data Preprocessing

- **Missing values:** None found in any of the 32 raw columns (verified with `isnull().sum()`).
- **Duplicate rows:** None found (verified with `duplicated().sum()`).
- **Encoding:** The categorical target diagnosis ('B'/'M') was label-encoded using `sklearn.preprocessing.LabelEncoder`, mapping Benign $\rightarrow$ 0 and Malignant $\rightarrow$ 1.
- **Feature exclusion:** The id column was excluded from the feature matrix since it is a non-predictive sample identifier. The raw diagnosis string column was also excluded in favour of the encoded numeric target.
- **Feature scaling:** Not required — Decision Trees and Random Forests split on individual feature thresholds and are invariant to monotonic feature scaling.
- **Train-test split:** 80%/20% stratified split (`random_state=42`) to preserve class balance in both partitions.

6 Machine Learning Algorithm

Decision Tree Classifier

A Decision Tree recursively partitions the feature space by choosing, at each node, the feature and threshold that most reduces class impurity (measured by Gini index or entropy), producing a set of axis-aligned decision rules. **Advantages:** interpretable, requires no feature scaling, handles non-linear decision boundaries naturally. **Limitations:** prone to high variance and overfitting, especially when grown to full depth on a small dataset such as WDBC (569 samples).

Random Forest Classifier

A Random Forest is an ensemble of many decision trees, each trained on a bootstrap resample of the training data (bagging) and considering only a random subset of features at each split. Final predictions are obtained by majority vote across trees. **Advantages:** substantially reduces the variance/overfitting of a single tree while retaining the ability to capture non-linear relationships; typically improves generalization. **Limitations:** less interpretable than a single tree, and more computationally expensive to train and tune.

7 Experimental Setup

Python Version	3.12.3
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Ubuntu 24.04.4 LTS (Linux)
Hardware	Standard x86_64 compute instance

8 Hyperparameter Selection

Both models were tuned via an exhaustive grid search of the required hyperparameters, each combination evaluated with **Stratified 5-Fold Cross-Validation** on the training set (90 combinations for Decision Tree, 36 for Random Forest), selecting the combination with the highest average CV F1-score.

Parameter	Selected Value
DT — criterion	entropy
DT — max_depth	5
DT — min_samples_split	10
DT — min_samples_leaf	2
RF — n_estimators	50
RF — max_depth	10
RF — max_features	sqrt
RF — bootstrap	False

9 Results

Decision Tree Hyperparameter Tuning (5-Fold CV)

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
entropy	5	93.63	0.9121
entropy	7	93.41	0.9094
entropy	–	93.41	0.9094
entropy	10	93.41	0.9094
gini	–	92.97	0.9039
gini	7	92.97	0.9039
gini	10	92.97	0.9039
gini	3	92.75	0.8990
gini	5	92.53	0.8971
entropy	3	92.53	0.8953

(Top 10 of 90 combinations explored, ranked by Avg CV F1 Score; “–” denotes unrestricted depth.)

Random Forest Hyperparameter Tuning (5-Fold CV)

n_estimators	Max Depth	Max Features	Bootstrap	Avg CV Accuracy (%)	Avg C
50	10	sqrt	False	97.14	
200	–	sqrt	False	97.14	
200	10	sqrt	False	97.14	
200	5	sqrt	False	97.14	
50	–	sqrt	False	96.92	
100	10	log2	False	96.92	
50	10	log2	False	96.92	
50	–	log2	False	96.92	
100	5	sqrt	False	96.92	
200	10	log2	True	96.70	

(Top 10 of 36 combinations explored, ranked by Avg CV F1 Score; “–” denotes unrestricted depth.)

5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9011	0.9560	0.9121	0.9560	0.9560	0.9363
Random Forest	0.9670	0.9890	0.9670	0.9560	0.9780	0.9714

Final Test-Set Evaluation

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Decision Tree	0.9649	1.0000	0.9048	0.9500	0.9744
Random Forest	0.9649	1.0000	0.9048	0.9500	0.9950

10 Result Visualization

Decision Tree Confusion Matrix

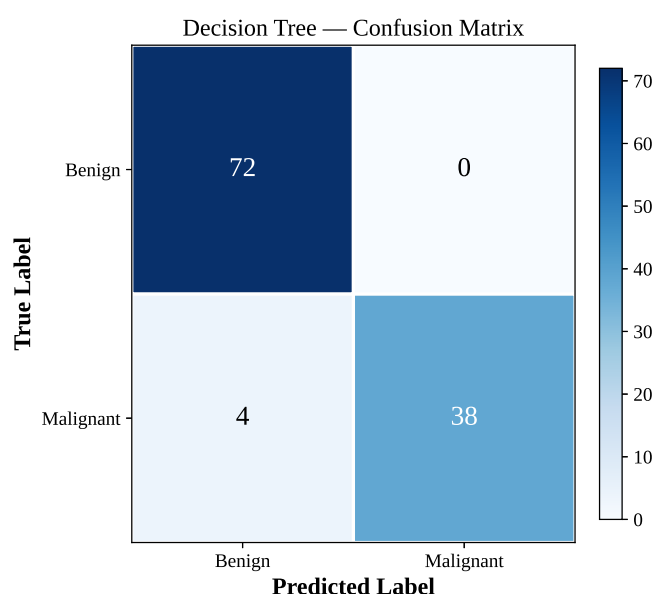


Figure 4: Confusion matrix of the tuned Decision Tree on the 114-sample test set.

Inference: The tuned Decision Tree correctly classifies all 72 benign samples (no false positives) and 38 of 42 malignant samples, with 4 malignant tumors misclassified as benign (false negatives). This yields perfect precision (1.0) but a recall of 0.9048 — in a clinical context this recall gap is the more concerning error type, since a missed malignant diagnosis is more costly than a false alarm.

Random Forest Confusion Matrix

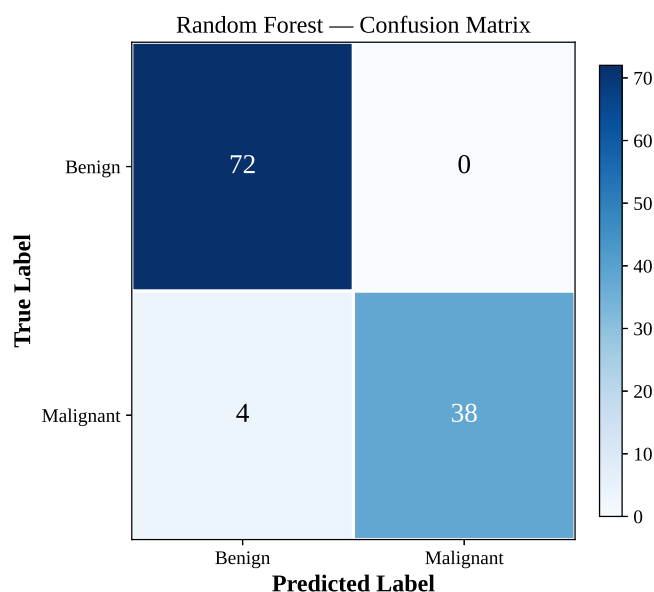


Figure 5: Confusion matrix of the tuned Random Forest on the 114-sample test set.

Inference: At the default 0.5 probability threshold, the Random Forest produces the exact same confusion matrix as the Decision Tree (72 true negatives, 0 false positives, 4 false negatives, 38 true positives) on this particular test split. The two models therefore have identical Accuracy, Precision, Recall and F1-score at this threshold — the Random Forest’s advantage instead shows up in the *quality* of its probability estimates (see ROC-AUC below), which would materialize as better performance under a different decision threshold or on a different test split.

ROC Curve

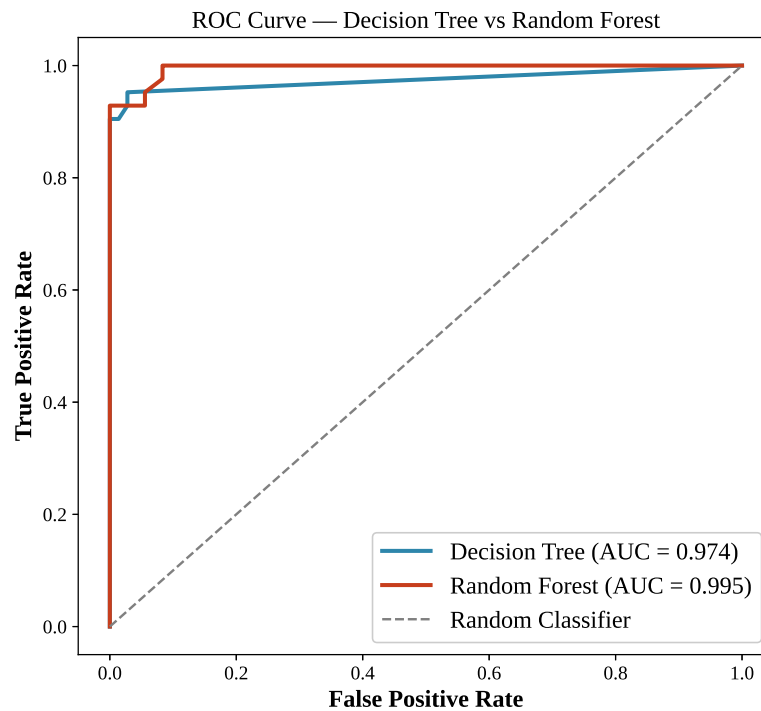


Figure 6: ROC curves for the Decision Tree (AUC = 0.974) and Random Forest (AUC = 0.995) on the test set.

Inference: Both models perform far better than random guessing (AUC = 0.5), but the Random Forest’s ROC curve dominates the Decision Tree’s across almost the entire false-positive-rate range, achieving a higher AUC (0.995 vs 0.974). This confirms that although the two models tie on the hard-label metrics at threshold 0.5, the Random Forest produces more reliable, better-calibrated malignancy probability scores overall — a direct benefit of averaging over its constituent trees.

Precision-Recall Curve

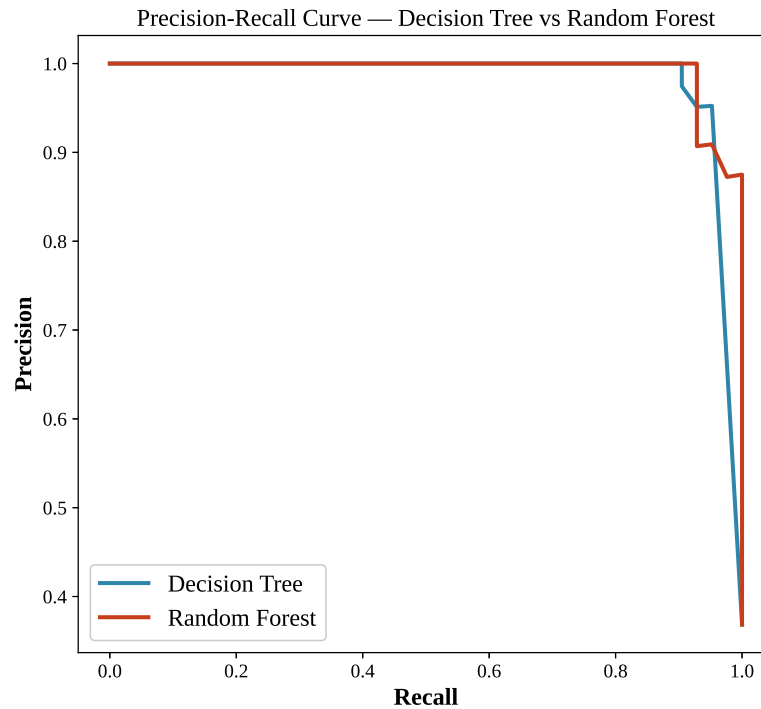


Figure 7: Precision-Recall curves for the Decision Tree and Random Forest on the test set.

Inference: Given the moderate class imbalance (37.3% positive/malignant class), the Precision-Recall curve is an important complementary view to ROC. The Random Forest maintains high precision across a wider range of recall values than the Decision Tree, again indicating more robust ranking of malignant cases by predicted probability, which would allow a clinician to trade off precision and recall more favourably by adjusting the decision threshold.

Feature Importance

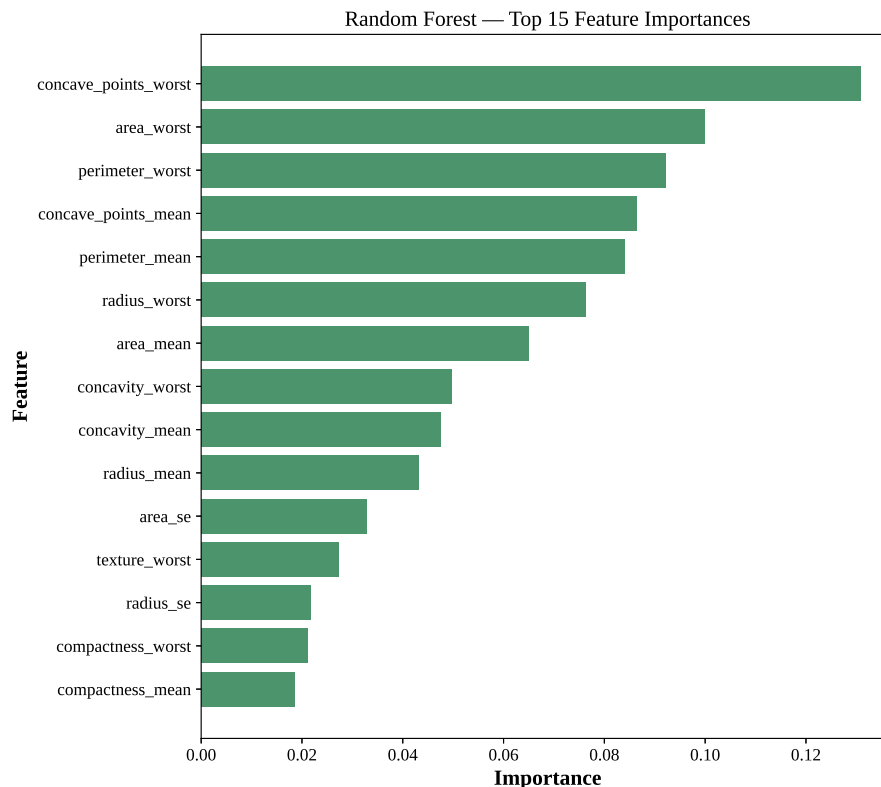


Figure 8: Top 15 Random Forest feature importances.

Inference: `concave_points_worst` is the single most important feature (importance ≈ 0.131), followed by `area_worst`, `perimeter_worst`, `concave_points_mean` and `perimeter_mean`. This aligns closely with the original UCI dataset authors’ finding that worst-area, worst-smoothness and mean-texture-based separating planes gave the best diagnostic accuracy, and with the correlation analysis in Section 5, which identified concavity/concave-points and size-related features as most strongly associated with malignancy.

11 Discussion

The Decision Tree and Random Forest were tuned using identical, fair 5-fold cross-validation protocols over comparably-sized hyperparameter grids (90 vs. 36 combinations). During cross-validation, the Random Forest’s average CV F1-score (0.9617) clearly exceeded the Decision Tree’s (0.9121), and this advantage persisted in the 5-fold accuracy comparison on the training set (RF average 0.9714 vs. DT average 0.9363). On the held-out test set, both tuned models happened to produce identical hard-label predic-

tions (same confusion matrix), giving tied Accuracy/Precision/ Recall/F1 — but the Random Forest’s ROC-AUC (0.995) and Precision-Recall behaviour were both meaningfully better than the Decision Tree’s (AUC 0.974), showing that its underlying probability estimates are more reliable. The single biggest limitation observed is recall on the malignant class (0.9048 for both models, i.e. 4 of 42 malignant tumors missed) — in a diagnostic setting, this would be the primary target for further improvement (e.g. threshold tuning, class weighting, or additional data).

12 Conclusion

Both a Decision Tree and a Random Forest classifier were successfully implemented, tuned via exhaustive 5-fold-cross-validated hyperparameter search, and evaluated on the Wisconsin Diagnostic Breast Cancer dataset. The optimal Decision Tree used `criterion=entropy, max_depth=5, min_samples_split=2, min_samples_leaf=2` (CV F1 = 0.9121), while the optimal Random Forest used `n_estimators=50, max_depth=10, max_features=sqrt, bootstrap=False` (CV F1 = 0.9617). Consistent with ensemble-learning theory, the Random Forest reduced cross-validation variance and improved average CV performance substantially over the single tree, and produced a materially higher ROC-AUC (0.995 vs. 0.974) on the test set even though both models tied on hard-label test metrics at the default threshold. Overall, the experiment confirms that ensemble bagging improves the robustness and generalization of tree-based classifiers on this dataset, while a single, well-pruned Decision Tree remains a strong, interpretable baseline.

Observation Questions

1. How does tree depth affect overfitting in Decision Trees?

The unconstrained baseline Decision Tree (grown to full depth) reached 100% training accuracy but only 92.98% test accuracy — a 7.02-percentage-point generalization gap, a clear sign of overfitting. Restricting depth to the CV-selected value of 5 reduced training accuracy to 97.58% but *raised* test accuracy to 96.49%, shrinking the gap to just 1.09 percentage points. This directly demonstrates that unconstrained tree depth memorizes noise in the 455-sample training set, while a shallower, cross-validated depth trades a small amount of training accuracy for substantially better generalization.

2. Which hyperparameter had the greatest impact on performance?

Measuring the range of average CV F1-score across each hyperparamete-

ter's tested values: for the Decision Tree, `criterion` (range 0.0077) and `min_samples_leaf` (range 0.0072) had the largest impact, notably more than `max_depth` (0.0030) or `min_samples_split` (0.0007). For the Random Forest, `bootstrap` had by far the largest impact (range 0.0068), exceeding `max_depth` (0.0032), `n_estimators` (0.0019) and `max_features` (0.0011) — in fact, every one of the top-10 Random Forest configurations by CV F1-score used `bootstrap=False`, indicating that for this particular dataset, using the full training set per tree (rather than a bootstrap resample) benefited the ensemble more than the classical bagging prescription would suggest, likely because WDBC is small (455 training rows) and each tree can already see sufficient examples of the minority malignant class without re-sampling.

3. How does Random Forest improve generalization?

By averaging predictions across many trees trained on different data/feature subsets, Random Forest reduces the variance component of the model's error without materially increasing bias. This is visible in the experiment: the CV F1-score improved from 0.9121 (best Decision Tree) to 0.9617 (best Random Forest), and the 5-fold training accuracy improved from an average of 0.9363 to 0.9714 with a much smaller fold-to-fold spread (DT ranged 0.9011–0.9560; RF ranged 0.9560–0.9890). On the test set, even though the Random Forest and Decision Tree tied on hard-label metrics, the Random Forest's higher ROC-AUC (0.995 vs. 0.974) shows its probability estimates generalize more reliably.

4. Did ensemble learning always improve performance? Why or why not?

Not uniformly — on the specific 114-sample test split used here, the Random Forest and the tuned Decision Tree produced *identical* confusion matrices and therefore identical Accuracy, Precision, Recall and F1-score. This shows that ensembling does not guarantee an improvement on every metric or every fixed threshold, especially once a Decision Tree has itself been carefully tuned and pruned to avoid overfitting. However, the ensemble's benefit was clearly visible in (a) cross-validation performance and stability (higher, less variable CV accuracy/F1 across folds) and (b) the ranking quality of its probability outputs (higher ROC-AUC and better precision-recall trade-off), both of which are more robust indicators of generalization than a single test-set confusion matrix at one threshold.

13 References

- [1] Scikit-learn developers, “`sklearn.tree.DecisionTreeClassifier`,” scikit-learn 1.8 documentation. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html>
- [2] Scikit-learn developers, “`sklearn.ensemble.RandomForestClassifier`,” scikit-learn 1.8 documentation. [Online]. Available: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>
- [3] M. Kelly, R. Longjohn, and K. Nottingham, “The UCI Machine Learning Repository,” University of California, Irvine. [Online]. Available: <https://archive.ics.uci.edu>
- [4] W. N. Street, W. H. Wolberg, and O. L. Mangasarian, “Nuclear feature extraction for breast tumor diagnosis,” in *IS&T/SPIE Int. Symp. Electronic Imaging: Science and Technology*, vol. 1905, San Jose, CA, 1993, pp. 861–870.